

Immunologic aspect of inflammatory bowel disease & celiac disease

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Inflammatory bowel disease (IBD)

- An immune-mediated chronic, progressive intestinal inflammatory disease associated with relapsing and remitting course.
- Ulcerative colitis (UC) and Crohn's disease (CD) are the two major types of IBD.
- The location of the inflammation, the disease's behavior, the symptoms, and the nature of the histological GI disorders differ between the two diseases.
- UC typically involves localized inflammation that begins in the rectum and extends proximally in a continuous manner to involve the entire colon in some cases.
- CD can affect any part of the GIT and is characterized by skip lesions.
- If progresses, it can lead to severe complications such as bowel obstructions or perforations, fistulas, and an increased risk of colorectal cancer in older patients.

- **Extra-intestinal manifestations** are common in both diseases including seronegative arthritis, uveitis, and sclerosing cholangitis...etc.
- **Malnutrition** is a frequent complication, resulting from reduced nutrient intake, malabsorption, and increased metabolic demands due to chronic inflammation.
- Although it can affect people of all ages, the **prevalence** of IBD is highest in the second to third decade of life and a second peak in the elderly population.
- **IBD** is believed to **result** from **a complex interplay between** genetic background, mucosal barrier dysfunction, disturbances in the GI microbiota (Dysbiosis), dysregulated immune responses, and environmental factors such as diet, air pollution, smoking, drugs, geography, and psychosocial stress.

In IBD suppression of inflammation (gut tolerance) is altered, leading to uncontrolled inflammation.

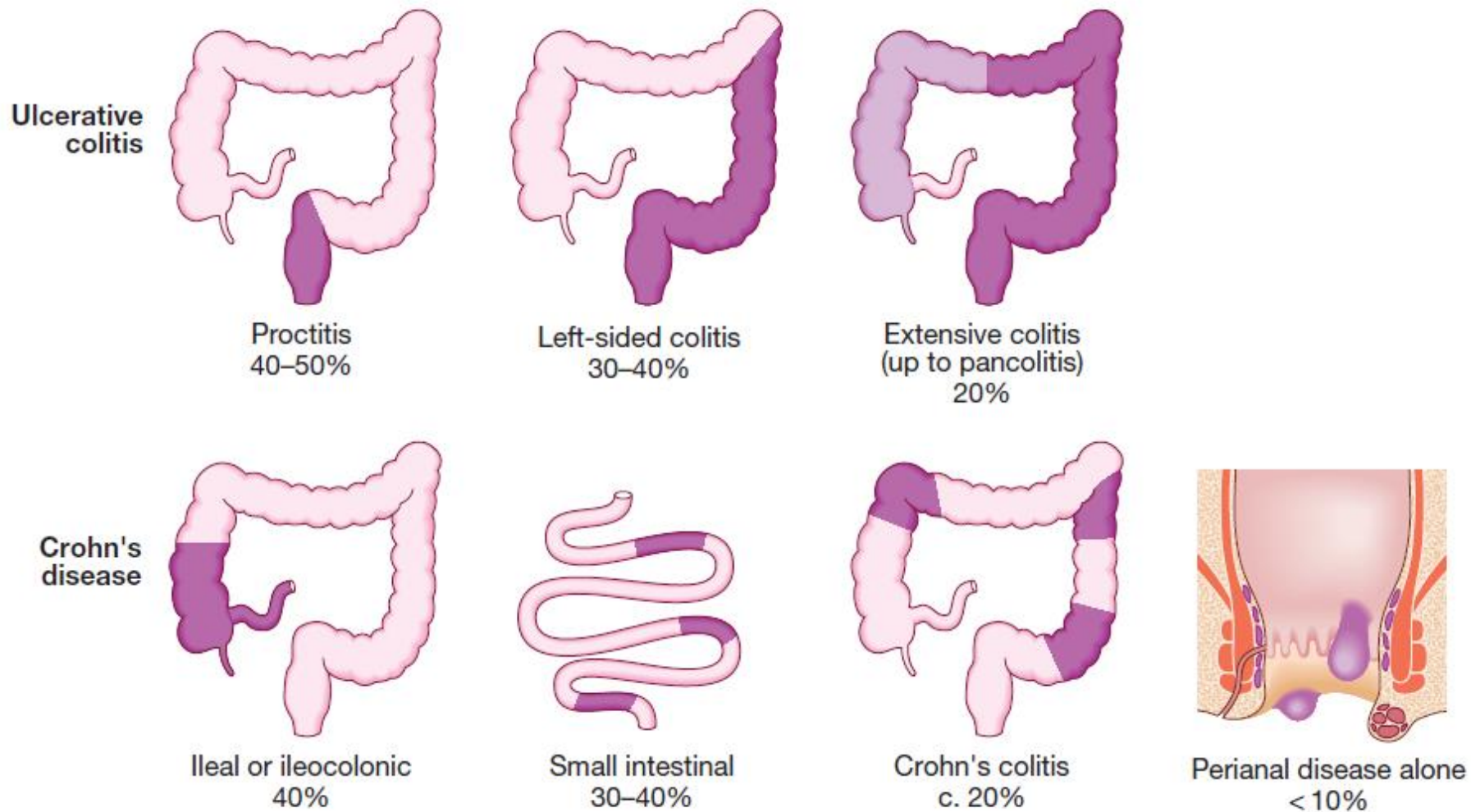
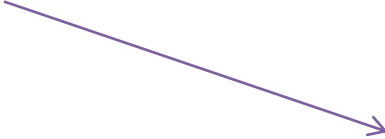


Fig. 21.45 Common patterns of disease distribution in inflammatory bowel disease.

- During the course of infections or other environmental stimuli in the **normal host**, full activation of the lymphoid tissues in the intestines occurs but is rapidly superseded by dampening of the immune response and tissue repair, however, **in IBD** such processes may not be regulated normally.
- **IBD** is considered to be a **polygenic disorder** that gives rise to multiple clinical subgroups within UC and CD.
- **First-degree relatives** have higher risk of developing IBD and the greater concordance in **homozygous twins**.
- **Genetically**, it is found to link with **mutations in genes encoding**, such as **IL-10, IL-10R, IL-23R....etc.**
- **Mutations** in the **CARD15** also known as **NOD2 gene**, encoding a cytoplasmic pattern recognition receptor for the muramyl dipeptide of bacterial cell wall peptidoglycan, are strongly associated with susceptibility to **CD**.

- HLA-DR*103 is associated with severe UC which is also associated with an increase in extra-intestinal complications.
- CARD9 expressed in neutrophils is involved in the protective role against colitis, the deficiency of which induces mitochondrial dysfunction with increasing production of ROS, leading to neutrophil apoptosis.
- Among the most relevant miRNAs found to be overexpressed in IBD in various studies are miR-21, miR-223, and miR-155, implicating them in the etiology and development of the disease.
- Active CD patients tend to adopt unbalanced nutritional patterns, while healthy eating patterns are associated with a lower risk of older-onset.
- Mediterranean diets, rich in polyphenols and fiber, are considered a beneficial dietary regimen for IBD.

- High consumption of fats and sugars and a low fiber intake, contribute to the development of IBD, because of the ↓ abundance of a short-chain fatty acid (SCFA) producing bacteria and the ↑ abundance of pathogens.



Patients with IBD patients show a ↓ in the number of bacterial species that produce propionate, acetate, and butyrate, a SCFA that positively modulates intestinal balance and reduces inflammation.

- The diagnosis is based on a combination of clinical, immunological, endoscopic, and histological findings.

The inflammatory cascade in IBD

- ✓ During the process of IBD, the physical barrier of the intestinal mucosa can deteriorate, consequently affecting both innate and adaptive immunity.
- ✓ Activation of innate immune cells, such as macrophages can result in the production of ROS during metabolic processes, leading to significant tissue damage.
- ✓ The activation of nuclear factor κ B (NF- κ B) and other signaling pathways results in the uncontrolled release of inflammatory cytokines during the IBD process and contributes to the inflammatory cascade reaction, the proliferation of memory T cells, and alterations in the intestinal microenvironment.
- ✓ Inflammatory cytokines from innate immune cells such as IL-1, IL-6, and TNF have diverse effects on tissues; they promote fibrogenesis, collagen production, activation of tissue metalloproteinases, and the production of other inflammatory mediators...etc.

- ✓ This **resulting** in an imbalance between the **pro-inflammatory** and **anti-inflammatory mediators**.
- ✓ Simultaneously, the imbalance between inflammatory Th cells and regT cells are essential for the progression of IBD.
- ✓ **↑** number of **Th17** cells in the peripheral blood of IBD patients, while **↓** number of **Treg cells**.
- ✓ **Th17** cells and **their cytokines** are important in **driving inflammation** in IBD.
- ✓ **Th17** cells may be **responsible for neutrophilic recruitment**.
- ✓ The **IL-23/Th17 pathway** also affects the progression of IBD.
- ✓ **↑infiltration of eosinophils** and **neutrophils** is often associated with more severe inflammation in IBD.

IL-23/Th17 pathway is particularly important in chronic intestinal inflammation, and abnormalities in IL-23 signaling have been implicated in both CD and UC.

- ✓ Additionally, **Innate Lymphoid Cells group 3 (ILC3s)** serving as the major mediator of chronic intestinal inflammation.
- ✓ **B-cell** perturbations have been observed, which may be related to the release of IgM, and other Igs.
- ✓ Enhanced **IgA** was produced in IBD patients against the **microbiota**.
- ✓ The **diagnosis** is based on a combination of clinical, immunological, endoscopic, and histological findings.
- ✓ In patients with IBD, **elevated CRP** levels have been associated with more severe histological phenotypes of the disease.
- ✓ **Elevated levels of lactoferrin** have been observed in patients with IBD compared to healthy individuals.
- ✓ **Elevated fecal lactoferrin** is a highly sensitive and specific marker for detecting intestinal inflammation of IBD from irritable bowel syndrome (IBS).

- ✓ Also, **calprotectin** can successfully **differentiate** between active and inactive IBD and providing a good monitoring tool.
- ✓ These characteristics of the immune response in IBD explain the **beneficial therapeutic effects** of Abs to block pro-inflammatory cytokines or the signaling by their receptors (e.g., anti-TNF, anti-IL-12, anti-IL-23, anti-IL-6) or molecules associated with leukocyte recruitment (e.g., anti- $\alpha 4\beta 7$ which is integrin receptor antagonists).
- ✓ **Therapies** such as the 5-aminosalicylic acid (5-ASA) compounds and corticosteroids are also potent inhibitors of these inflammatory mediators through inhibition of transcription factors such as NF- κ B that regulate their expression.
- ✓ **Probiotics, prebiotics, and fecal microbiota transplantation** are modern techniques for managing IBD.

Ulcerative colitis

- ❖ In severe disease, the mucosa is **hemorrhagic** (very common), edematous, and ulcerated.
- ❖ In long-standing disease, **inflammatory polyps** (pseudopolyps) may be present as a result of epithelial regeneration.
- ❖ Patients with fulminant disease can develop a **toxic colitis or megacolon** where the bowel wall thins and the mucosa is severely ulcerated; this may lead to perforation.
- ❖ **The major symptoms** of UC are diarrhea, rectal bleeding, tenesmus, passage of mucus, and crampy abdominal pain.
- ❖ Both **acute** and **chronic inflammatory cells** infiltrate the lamina propria and the **cryptitis** and **crypt abscesses** are typical.
- ❖ UC is more common in **non-smokers** and **ex-smokers**.

Smoking described as a risk factor for CD and, surprisingly, a protective factor for UC.

- ❖ Emotional stress, intercurrent infection, gastroenteritis, antibiotics or NSAID therapy may all **provoke a relapse**.
- ❖ **Active disease** can be associated with a rise in CRP, platelet count, ESR, and a decrease in hemoglobin.
- ❖ **Increased** levels of perinuclear antineutrophil cytoplasmic Abs (**pANCA**) are more commonly seen in patients with **UC**.
- ❖ **pANCA** positivity is associated with pancolitis and primary sclerosing cholangitis.
- ❖ 10– 15% may have anti- *Saccharomyces cerevisiae* Abs (**ASCA**).

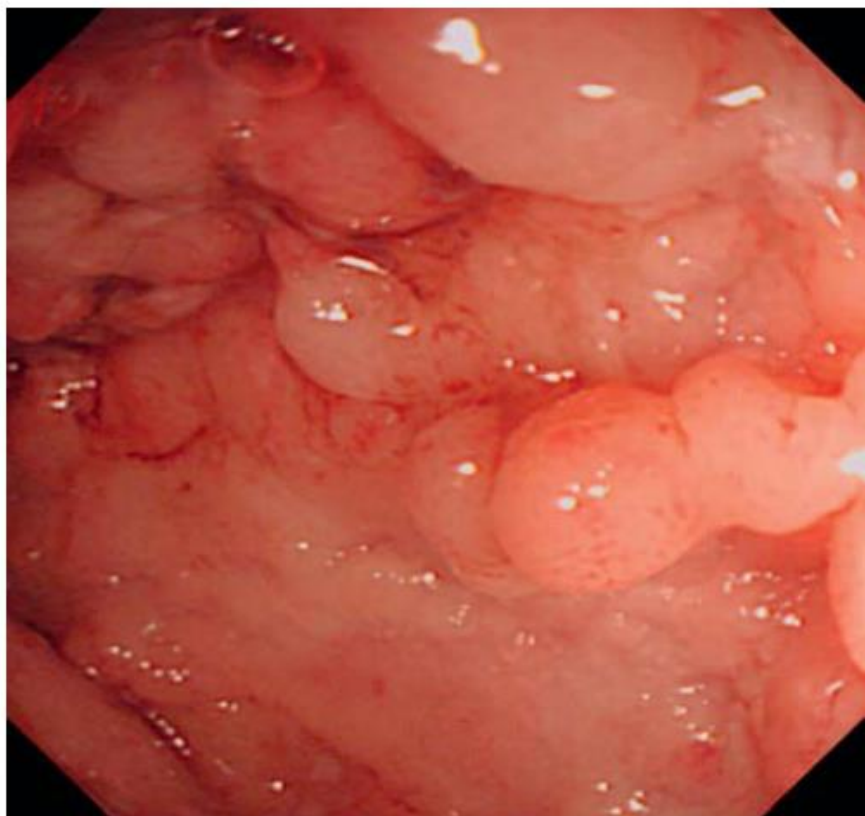


Fig. 21.46 Pseudopolyposis in ulcerative colitis.



FIGURE 319-7 Colonoscopy with acute ulcerative colitis: severe colon inflammation with erythema, friability, and exudates. (Courtesy of Dr. M. Hamilton, Gastroenterology Division, Department of Medicine, Brigham and Women's Hospital, Boston, Massachusetts; with permission.)

Crohn's disease (CD):

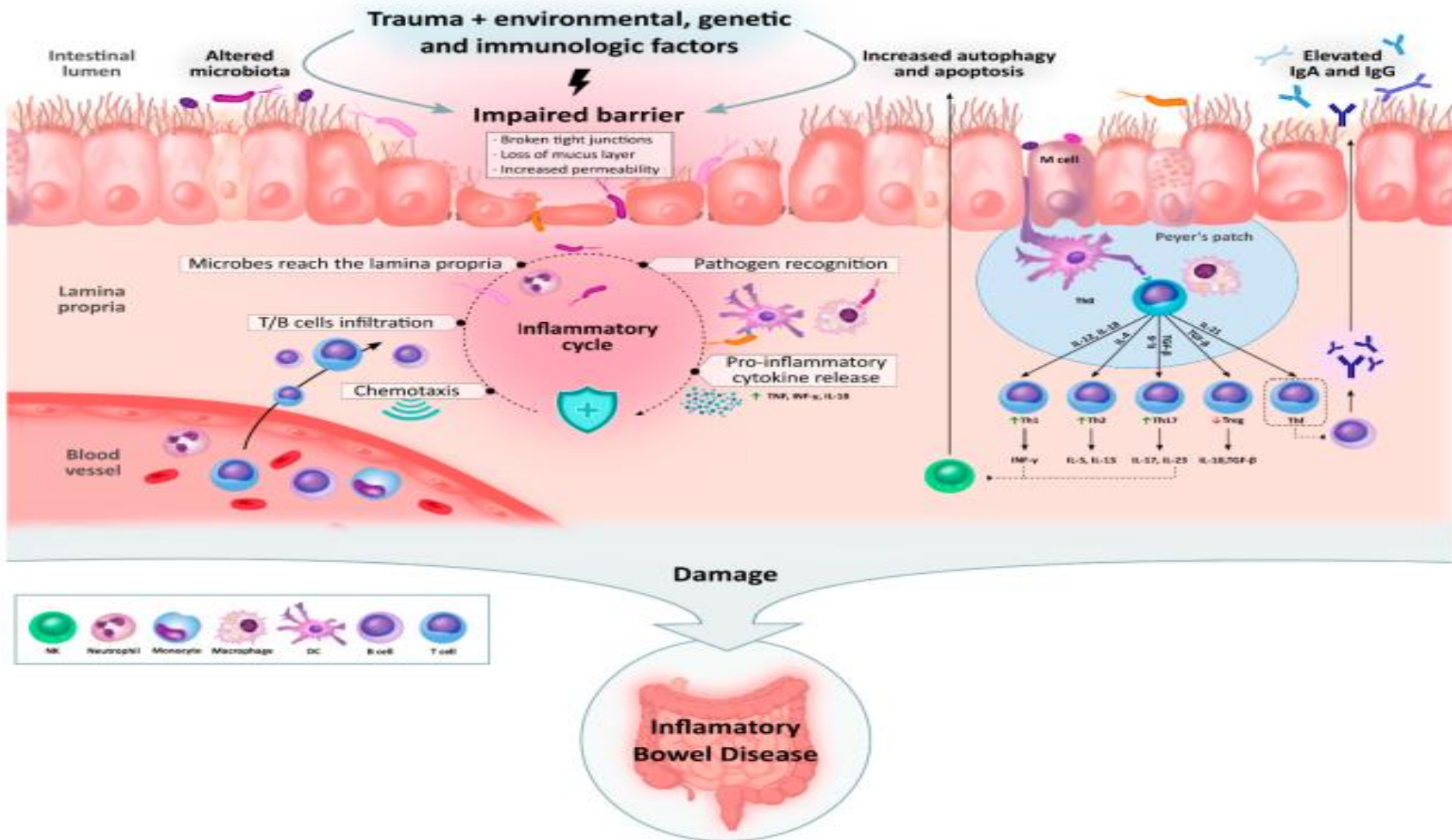
- The **etiology** is not known but it has been suggested that it is an **autoimmune disease** (mostly immune mediated disease).
- There is evidence of both a **cell mediated response** and **defective T cell suppressor function**.
- **CD** is associated with specific **auto-Abs**.
- Patients produce **Abs** to a variety of organisms, and 60-70% have Abs to the yeast *Saccharomyces cerevisiae* (**ASCA**).
- **pANCA** in Crohn's is associated with colonic disease.
- **Other Abs** found in CD including anti cardiolipin, rheumatoid factors, anti goblet cell...etc.
- **CD** is more common in smokers.
- **CD** is associated with a low-residue, high-refined-sugar diet.

- **Active CD** is characterized by focal inflammation and formation of fistula tracts, which resolve by fibrosis and stricturing of the bowel.
- **Granulomas** are a characteristic feature of CD.
- **Extensive inflammatory disease** is associated with a loss of digestive and absorptive surface, resulting in **malabsorption**.
- The major **symptoms** are abdominal pain, diarrhoea and weight loss.
- **Laboratory abnormalities** include elevated ESR and CRP.
- In more **severe disease**, findings include hypoalbuminemia, anemia, and leukocytosis.
- **Perforation** occurs in 1–2% of patients, usually in the ileum but occasionally in the jejunum or as a complication of toxic megacolon.

Increased titers of ASCAs have been associated with CD, whereas increased levels of pANCA are more commonly seen in patients with UC.



FIGURE 319-8 Wireless capsule endoscopy image in a patient with Crohn's disease of the ileum shows ulcerations and narrowing of the intestinal lumen. (Courtesy of Dr. S. Reddy, Gastroenterology Division, Department of Medicine, Brigham and Women's Hospital, Boston, Massachusetts; with permission.)



Overview of key inflammatory events initiating inflammation. Initially, environmental, genetic, and immunological factors disrupt the intestinal epithelial barrier, allowing microbiota entry into the lamina propria and triggering an inflammatory cascade. Dendritic cells (DCs) and macrophages recognize pathogens, presenting antigens to T and B cells, resulting in cytokine release. In Peyer's patch, a gut-associated lymphoid tissue (GALT), DCs stimulate Th0 cell differentiation into specific T cell subtypes based on environmental cytokines. B cells produce elevated levels of immunoglobulins (Igs) A and G in the intestinal lumen, while natural killer (NK) cells promote autophagy and apoptosis.

Coeliac disease

- Celiac disease has had several other names, including non-tropical sprue, celiac sprue, adult celiac disease, and gluten-sensitive enteropathy.
- It is an autoimmune mediated a chronic inflammatory disorder of the small bowel occurring in genetically susceptible individuals, which results from intolerance to wheat gliadin (gluten) and similar proteins found in rye, barley and, to a lesser extent, oats.
- The condition occurs worldwide and can present at any age.
- First-degree relatives and monozygotic twins are at high risk.
- Celiac disease is considered an “iceberg” disease.

Pathophysiology

- The precise mechanism of mucosal damage is unclear but immunological responses (both adaptive and innate immune responses) to gluten play a key role.
- inflammatory reaction in the small bowel mucosa leading to villous atrophy with crypt hyperplasia in genetically predisposed people.
- There is a strong association HLA-DQ2/DQ8 allele.
- Dysbiosis of the intestinal microbiota has been identified but it is unclear if this is pathological or a response to the underlying mucosal changes.
- The disorder involves what is probably a genetically related increased mucosal activity of transglutaminase (the main target Ag of anti-endomysium Abs).
- B cells produce Abs against both tissue transglutaminase and gliadin and deamidated gliadin peptides.

- This enzyme deamidates the glutamine residues in gliadin and creates a new T-cell epitope that binds efficiently to DQ2 and is recognized by IFN γ -secreting intraepithelial CD4⁺ Th1-cells that cause tissue injury.
- Local production of IL-15 also plays a role by increasing expression of non-classical MHC class I molecules on epithelial cells and receptors for these on intraepithelial CD8⁺ T-cells, $\gamma\delta$ T-cells, and NK cells, leading to cytotoxic killing of the epithelial cells.
- Serum Abs— IgA antigliadin, antiendomysial, and anti-tTG Abs are present.
- Patients with these Abs should undergo duodenal biopsy.
- Celiac disease is also associated with other diseases like diabetes mellitus type 1, IgA deficiency, Down's syndrome, Addison's disease, and thyroid disease...etc.

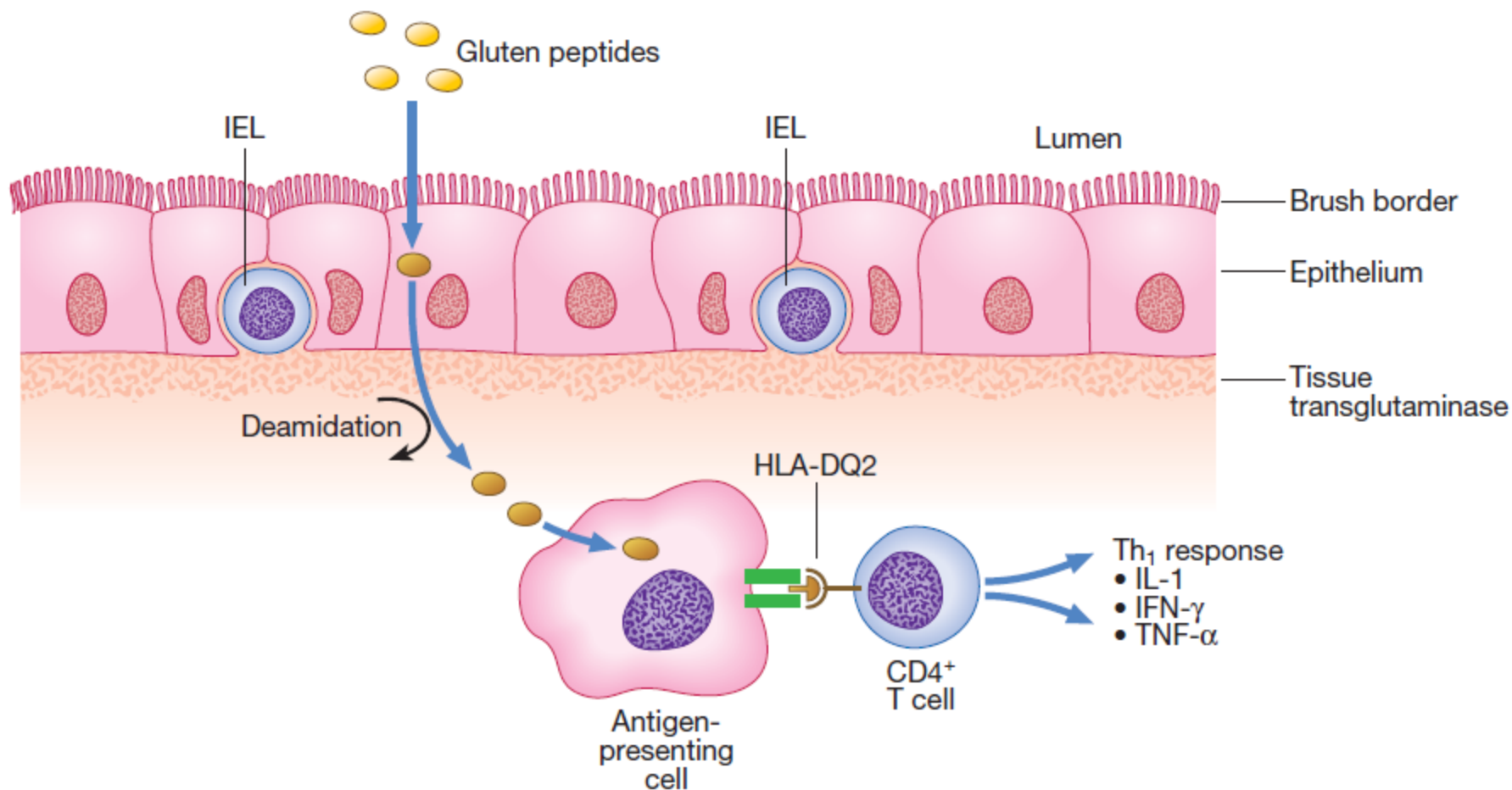


Fig. 21.41 Pathophysiology of coeliac disease. After being taken up by epithelial cells, gluten peptides are deamidated by the enzyme tissue transglutaminase in the subepithelial layer. They are then able to fit the antigen-binding motif on human leucocyte antigen (HLA)-DQ2-positive antigen-presenting cells. Recognition by CD4⁺ T cells triggers a Th₁ immune response with generation of pro-inflammatory cytokines: interleukin-1 (IL-1), interferon gamma (IFN-γ) and tumour necrosis factor alpha (TNF-α). Lymphocytes infiltrate the lamina propria and an increase in intra-epithelial lymphocytes (IELs), crypt hyperplasia and villous atrophy ensue.

Clinical features

- ✓ In **infancy**, it occurs after weaning on to cereals and typically presents with diarrhoea, malabsorption and failure to thrive.
- ✓ In **older children**, it may present with non-specific features, such as delayed growth, and delayed puberty.
- ✓ In **adults**, the disease usually presents during the third or fourth decade and females are affected twice as often as males.
- ✓ The **presentation** is **highly variable**, depending on the severity and extent of small bowel involvement.

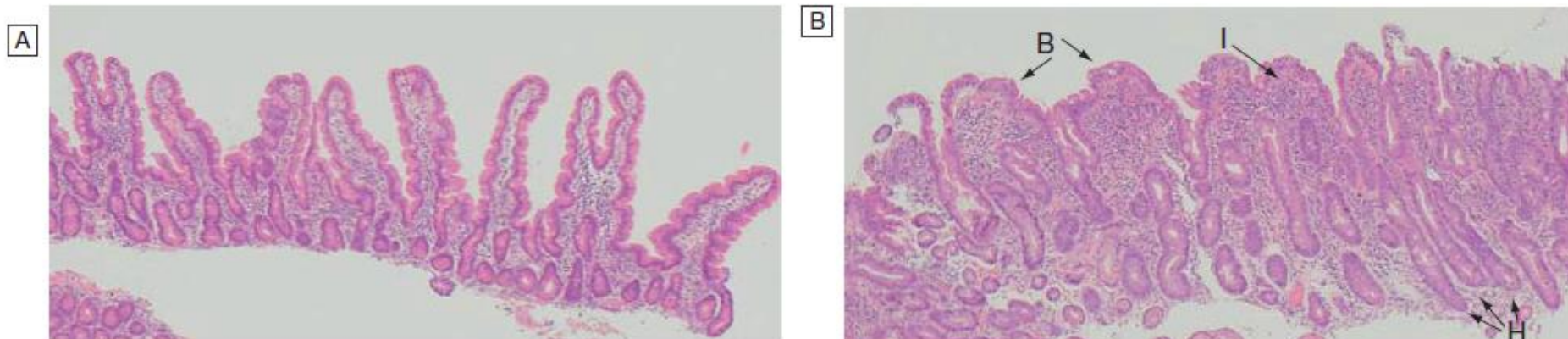


Fig. 21.42 Jejunal mucosa. **A** Normal. **B** Subtotal villous atrophy in coeliac disease. There is blunting of villi (B), crypt hyperplasia (H) and inflammatory infiltration of the lamina propria (I).

- ✓ A small number of individuals have **classic symptoms** (diarrhea, abdominal pain, vomiting, and fatty stools), and manifestations related to nutrient malabsorption.
- ✓ A much larger number of individuals have “**atypical celiac disease**,” with manifestations that are not obviously related to intestine symptoms (e.g., **anemia, osteopenia, infertility, and neurologic symptoms**).
- ✓ Finally, an even larger number of persons have “**silent celiac disease**”; they are essentially asymptomatic despite abnormal small-intestinal histopathology and serologies.
- ✓ In many patients, **frequent spontaneous remissions and exacerbations occur**.

- A twofold-increased risk of **malignancy**, particularly of enteropathy associated T-cell lymphoma.
- A few patients develop **ulcerative jejuno-ileitis**.

Investigations and Diagnosis

- ❑ The **gold standard** for diagnosis is represented by the combination of mucosal changes detected by duodenal biopsy and by positivity of serological tests (antitTG Abs, anti-endomysium Abs (EmA), and deamidated gliadin peptide (DGP) Abs).
- ❑ A **biopsy** should be performed when patients have symptoms and laboratory findings suggestive of nutrient malabsorption and/or deficiency as well as a positive tTG Ab test.
- ❑ The **histological features** are usually characteristic but other causes of villous atrophy, crypt hyperplasia, lymphocytic infiltration should be considered.
- ❑ **Antibody tests** constitute a valuable screening tool in patients with diarrhoea or other suggestive symptoms.
- ❑ **High-titre serology in children** can be diagnostic without the need for endoscopy and biopsy.

- ❑ Ab titers usually become negative with successful treatment.
- ❑ Anti-endomysial Abs of the IgA class are standard serological tests for coeliac disease.
- ❑ The determination of total IgA is necessary due to its frequent deficiency in CD patients. As a result, low levels of IgA anti-tTG Abs may result from total IgA deficiency and produce a false-negative result.
- ❑ IgG Abs must be analysed in patients with coexisting IgA deficiency.
- ❑ A full blood count may show microcytic or macrocytic anemia from iron or folate deficiency.
- ❑ There are lower levels of vitamin D among celiac disease and other fat soluble vitamins.

Treatment

- Treatment is lifelong **gluten avoidance**.
- **Abnormal bowel returns** to normal when gluten is withdrawn from the diet.
- **Abs gradually disappear** with gluten avoidance and reappear on gluten challenge.
- Nutritional Management such as iron, folate, vitamin B12, and vitamin D...etc.
- Follow up

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